

INTERNSHIP PROGRESS REPORT

Smt. Maherbanu College of Science & Commerce, Akola
Department of Computer Application (BCA)

1. STUDENT DETAILS

Particular	Details
Name of Student	Sushant Gangaprasad Ukande
Roll No	
Class	BCA 3 rd Year
Semester	Semester 5
Academic Year	2026–2027
Name of Organization	TechnoGrowth Software
Internship Domain	Machine Learning
Internship Duration	1 Months
Industry Mentor	
Faculty Mentor	Prof. Avani Kulkarni (H.O.D)

2. INTERNSHIP OBJECTIVES

Master Python Fundamentals:

Learn essential programming concepts like variables, loops, functions, and data structures using Python and Jupyter Notebook.

Clean and Analyze Data:

Use powerful libraries like NumPy and Pandas to organize datasets, clean up missing information, and perform basic statistical analysis like calculating averages and correlations.

Visualize Key Insights:

Build clear charts: including line graphs, bar charts, histograms, and scatter plots: using Matplotlib and Seaborn to spot trends and relationships in your data.

Build Your First Machine Learning Model:

Understand the basics of machine learning, distinguish between supervised and unsupervised learning, and train and evaluate a Linear Regression model using Scikit-Learn

3. WEEKLY INTERNSHIP PROGRESS

Week	Dates	Activities / Tasks Completed	Skill / Knowledge
Week 1		Python Basics for Data Science Introduction to Data Science Install tools • Installed Python and Jupyter Notebook. • Learned fundamental programming concepts and logic.	Python, Jupyter Notebook Python Basics: Variables, Data Types Operators Conditional Statements (if, else) Loops (for, while) Functions Lists, Tuples, Dictionaries
Week 2		Data Analysis with Python Introduction to NumPy Arrays • Performed mathematical operations on datasets. • Cleaned messy data, handled missing values, and grouped data.	Mathematical operations, Pandas Series and DataFrames Data cleaning Handling missing values Data filtering and grouping
Week 3		Data Visualization & Statistics • Created visual charts to understand data trends. • Calculated basic statistical metrics to analyze datasets.	Data visualization basics Charts: Line chart, Bar chart Histogram Scatter plot Visualization with Seaborn, statistics Mean, Median, Mode Standard Deviation Correlation
Week 4		Machine Learning Basics & Project • Learned core ML concepts and built a price prediction model. • Split datasets and tested model performance.	Introduction to Machine Learning Supervised vs Unsupervised Learning Linear Regression basics Model training using Scikit-learn Model evaluation

4. Project / Work Assigned

- **Project Title 1:** House Price Prediction System
- **Project Title 2:** Retail Sales Data Analysis & Reporting Dashboard

Technologies Used: Python, Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, Jupyter Notebook

5. Work Completed

- Cleaned and organized raw datasets by removing missing values and fixing incorrect entries.
- Analyzed sales data using Pandas to find top-selling products, regional performance, and total revenue.
- Built visual charts like bar graphs and scatter plots to easily see trends in housing prices and sales numbers.
- Built a Linear Regression model in Python to predict house prices based on size and number of bedrooms, testing it on unseen data to make sure it worked well.

6. Skills Acquired

1. Technical Skills

- Python for Data Science: Writing Python code using libraries like Pandas and NumPy to work with tables and numbers.
- Basic Machine Learning: Knowing how to train a model, split data into training and testing sets, and check accuracy.
- Data Visualization: Making clean graphs using Matplotlib and Seaborn to show findings visually.
- Data Cleaning: Fixing messy or incomplete data so it is ready for analysis.

2. Soft Skills

- Logical Thinking: Looking at numbers to find real-world meaning behind them.
- Problem-Solving: Fixing bugs in code and figuring out why a model isn't giving good results.
- Communication: Explaining complex data findings in a clear, simple way.

7. Challenges Faced and Solutions

Challenge	Solution
The machine learning model was just memorizing the training data instead of actually learning patterns.	Used <code>train_test_split</code> to separate the data, testing the model only on new, unseen data to keep it accurate.
The dataset had missing values and wrong formatting, which stopped the code from running.	Used simple Pandas commands like <code>fillna()</code> to clean up missing information before running any analysis.
It was tricky to figure out which features (like square feet or room count) actually affected house prices.	Created simple visual scatter plots to see which factors had the strongest connection to the final price.
Connecting math formulas to actual Python code felt confusing at first.	Broke down the concepts step-by-step and relied on Scikit-Learn's built-in functions to handle the heavy math.

8. Student Takeaways

- Gained hands-on experience using Python to solve real-world problems through data analysis and machine learning.
- Understood how to turn raw, messy data into clean tables and clear visual graphs.
- Learned how to train a basic predictive model from scratch and evaluate its performance.
- Built confidence in troubleshooting code errors and thinking critically about data patterns.

9. Future Plan / Next Steps

1. Advanced Machine Learning: Exploring algorithms like Decision Trees, Random Forests, and Logistic Regression for classification tasks.
2. Data Wrangling: Mastering complex SQL queries and advanced Pandas functions to handle larger, real-world datasets.
3. Interactive Dashboards: Learning tools like Power BI or Streamlit to present data visually for non-technical users.
4. Feature Engineering: Learning how to transform raw data attributes into better inputs to improve model accuracy.

10. Attendance

Particular	Details
Total Internship Duration	4 Weeks (28 Days)
Weekly Offs (Sundays)	4 Days
Total Working Days	28 Days
Days Attended	26 Days
Leave / Absence	2 Days
Attendance Percentage	92.86% (~93%)

11. Overall Progress

The student made great progress during the 4-week internship. They started with basic Python concepts and quickly moved on to analyzing real data and building machine learning models.

Throughout the project, they learned how to clean messy data, create clear graphs, and write working code to solve problems. They completed all tasks on time, handled coding errors well, and showed a strong interest in learning new skills. Overall, the student did a great job and gained a solid understanding of data science basics.

Signature of Mentor